

Linear Algebra — Concept Map

how we built from idea to idea

study notes

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1 The mind map



2 How we built: concept $\rightarrow$ concept

1. **Vector space** — the foundation. A set with addition and scalar multiplication (closure, zero vector, axioms). A vector is any element; in $\mathbb{R}^n$ it's a stack of numbers.
2. **Representation $\rightarrow$ dimension.** $\mathbb{R}^n$ has n directions, P_n has $n + 1$, P has $\aleph_0$, C is uncountable. Dimension = number of independent directions (basis size), not vector length.
3. **Free vs fixed.** Displacement vs position reading; the math runs on displacements (a basis, never an origin). This is why $[3, 5]$ needs no reference point as a displacement.
4. **Norm.** Add “size” to a vector space (3 axioms: positive definiteness, homogeneity, triangle inequality). Different norms = different geometry (unit spheres: circle, diamond, square, ellipse).
5. **Metric / distance.** Every norm induces a metric $d(x, y) = \|x - y\|$; “ $\|x\|$ = distance to the origin” is the position-reading corollary, not the definition.
6. **Triangle inequality.** Its engine is the projection split: $v = v_{\parallel} + v_{\perp}$, and $\|v_{\parallel}\| \leq \|v\|$ — a projection is never longer than the vector it came from.
7. **Cauchy–Schwarz.** That engine in dot-product clothes: $|\langle u, v \rangle| \leq \|u\| \|v\|$. Same fact, two languages.
8. **Inner product.** Add “alignment” between two vectors ($\langle x, y \rangle = x^T y$). It induces the norm $\|x\| = \sqrt{\langle x, x \rangle}$ and brings angles, orthogonality, and projections.
9. **Parallelogram law / polarization.** The bridge back: a norm comes from an inner product iff $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$ (Jordan–von Neumann), and polarization recovers the inner product from the norm — the norm was the inner product in disguise.
10. **Weighted inner products.** Any symmetric positive-definite M works ($\langle x, y \rangle_M = x^T M y$); the M norm is one of them. ℓ_2 is the flagship ($M = I$), the only rotation-invariant norm. That's the deep payoff: *the dot product induces the L2 norm, and L2 is the only norm invariant under rotation* — which is why L2's geometry matches everyday space, while the weighted norms are just L2 recoordinated.
11. **Span & independence.** $\text{span}(X)$ = the set of ALL linear combinations of X — everything buildable from X (always a subspace). A vector adds nothing if it's already in the span (redundancy). Independence is the formal test: $\sum c_i v_i = 0 \Rightarrow \text{all } c_i = 0$ (zero reachable only trivially). Independence $\Leftrightarrow$ no redundancy.
12. **Basis & dimension.** A basis = an independent spanning set = a minimal spanning set. *n independent vectors in $\mathbb{R}^n$ are automatically a basis* (independence + full count leaves nothing missing). $\dim(M)$ = number of vectors in ANY basis = an exact equality, not a bound.
13. **Determinants & Cramer.** $w = a_1 v_1 + \dots + a_n v_n$ is a system $A \cdot (\text{coeffs}) = w$; $\det A \neq 0 \Leftrightarrow$ invertible $\Leftrightarrow$ unique solution for every w . Cramer: each unknown is a ratio of determinants (swap in w 's column) — $\det$ is literally the denominator of the solve. $\det = 0 \Leftrightarrow$ dependent $\Leftrightarrow$ no unique solution. For big n : Gaussian elimination instead of hand substitution.
14. **Gram–Schmidt.** Manufactures an orthonormal basis of the SAME span: normalize the first; for each next vector subtract the projections onto ALL already-built u 's, then normalize the leftover. Perpendicularity is by construction — removing every parallel part leaves only the perpendicular part. In $\mathbb{R}^3$ the third direction is forced once u_1, u_2 are fixed ($u_3 = \pm u_1 \times u_2$).
15. **Orthonormal vs canonical basis.** Orthonormal $\Rightarrow$ coefficients are dot products $a_i = \langle w, u_i \rangle$ (read off, no solving); non-orthogonal $\Rightarrow$ coefficients are entangled (must solve a system). The canonical basis $e_1 \dots e_n$ is the DEFAULT orthonormal basis, where the coordinates ARE the vector's entries.

3 The ladder of structure

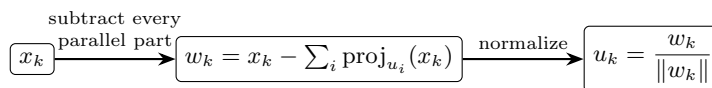
Every step adds structure; the richest sits at the top.

$$\begin{array}{ccccc} \text{inner product space} & \subset & \text{normed space} & \subset & \text{metric space} \\ \text{(sizes + angles)} & & \text{(sizes)} & & \text{(just distances)} \\ \text{vector space is the base: set + addition + scaling} & & & & \end{array}$$

- **vector space** $\rightarrow$ **normed space**: add $\|\cdot\|$ (3 axioms).
- **normed space** $\rightarrow$ **inner product space**: add $\langle \cdot, \cdot \rangle$, which induces the norm.
- **normed space** $\rightarrow$ **metric space**: every norm gives $d(x, y) = \|x - y\|$ (the triangle inequality is the norm's own axiom).
- Each step up is one-directional: inner product $\Rightarrow$ norm $\Rightarrow$ metric, but not back (except when the parallelogram law holds).
- **Coordinates ride on the basis, not on the space**: in an orthonormal basis, coefficients = dot products; in the canonical basis they are simply the entries themselves; in a non-orthogonal basis they need a system solve.

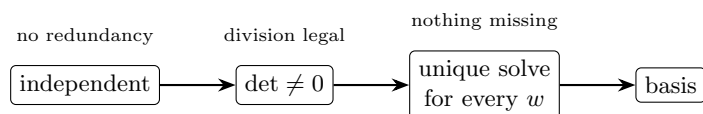
4 Gram–Schmidt and the basis chain, at a glance

4.1 The Gram–Schmidt loop



leftover w_k is $\perp$ to ALL of $u_1, \dots, u_{k-1}$ by construction: each projection removes one existing direction

4.2 The independence $\Leftrightarrow$ basis chain (n vectors in $\mathbb{R}^n$)



5 Gotchas worth remembering (where I got stuck)

- P_n has $n + 1$ **dimensions** (off-by-one: degree $\leq n$ means $n + 1$ coefficients); $\dim(P) = \aleph_0$. Infinity lives in the SPACE, never in one vector.
- **A dot product is always a scalar.** $(1, 2, 1) \cdot (1, 1, 0) = 3$, not $(1, 2, 0)$. The scalar then scales the WHOLE direction vector in $\text{proj}_a(v) = \frac{v \cdot a}{a \cdot a} a$.
- **Sum of projections $\neq$ projection of the sum.** $\text{proj}_{u_1}(v) + \text{proj}_{u_2}(v)$ is legal (axes kept separate); $\text{proj}_{u_1+u_2}(v)$ is wrong — $u_1 + u_2$ is ONE line = ONE perpendicularity, but covering two axes needs two.
- **A basis must be independent** (definition); orthogonality is the optional upgrade that makes coefficients free.

- **In a non-orthogonal basis, projection $\neq$ coefficient** (2.2 vs true $1/3$) — the axes overlap, so you must solve a system.
- **$\det = 0$ means the division is illegal** — no unique solution (none, or infinitely many).
- **Always check $\|u_k\| = 1$ after normalizing** — catches the “forgot to divide by the norm” slip.
- **Projection scalars are not coordinates.** In the 3D example, $\sqrt{2/3}$ was $v_3 \cdot u_2$ (a scalar) and u_2 ’s third coordinate, but NOT part of u_3 .